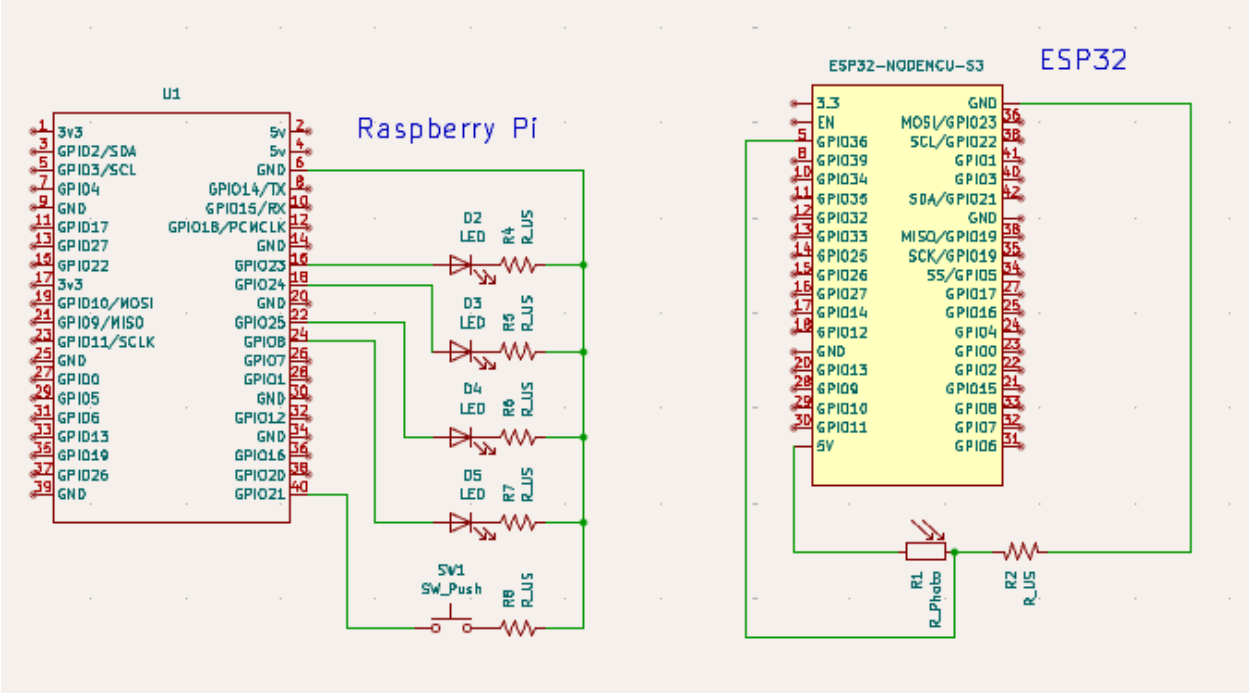
# We Need to Talk

Autumn Rogers

## Overview

### Schematic

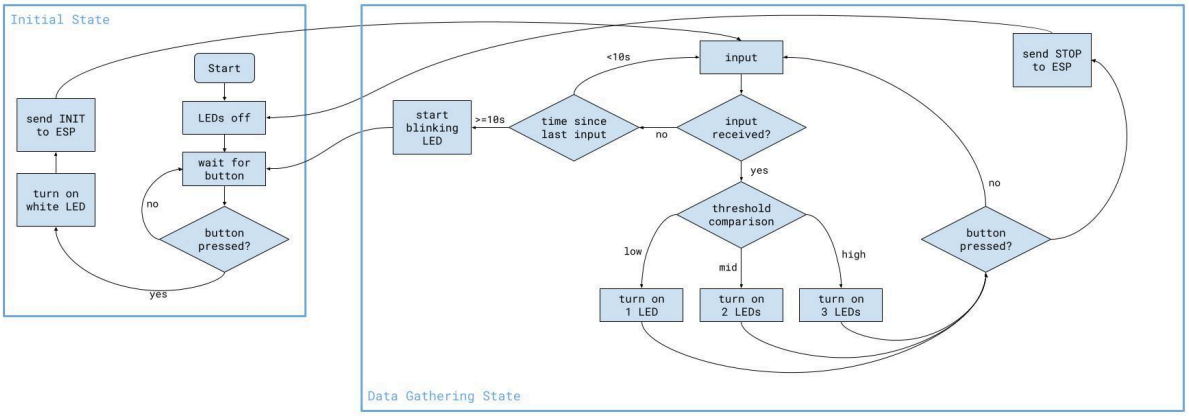



### Communication Protocol

| Sender | Receiver | Message | Purpose |
|---|---|---|---|
| Pi | ESP | `INIT` | Begin collecting data |
| Pi | ESP | `STOP` | Stop collecting data, return to ready state |
| ESP | Pi | `XXXX` | Integer representing the rounded average of the photoresistor data |
| Pi | ESP | `WTFDUDE` | Querying an ESP that has stopped sending data |
| ESP | Pi | `SORRY` | If this ESP receives WTFDUDE, it replies with SORRY to indicate that it's still active and ready to resume sending data |

# Raspberry Pi WiFi setup and packet delivery

## Flowchart




[Link to flowchart]

## Initial State

In the initial state, the Raspberry Pi turns off all LEDs and waits for the button press, which will initiate communication by sending an `INIT` message to the ESP.

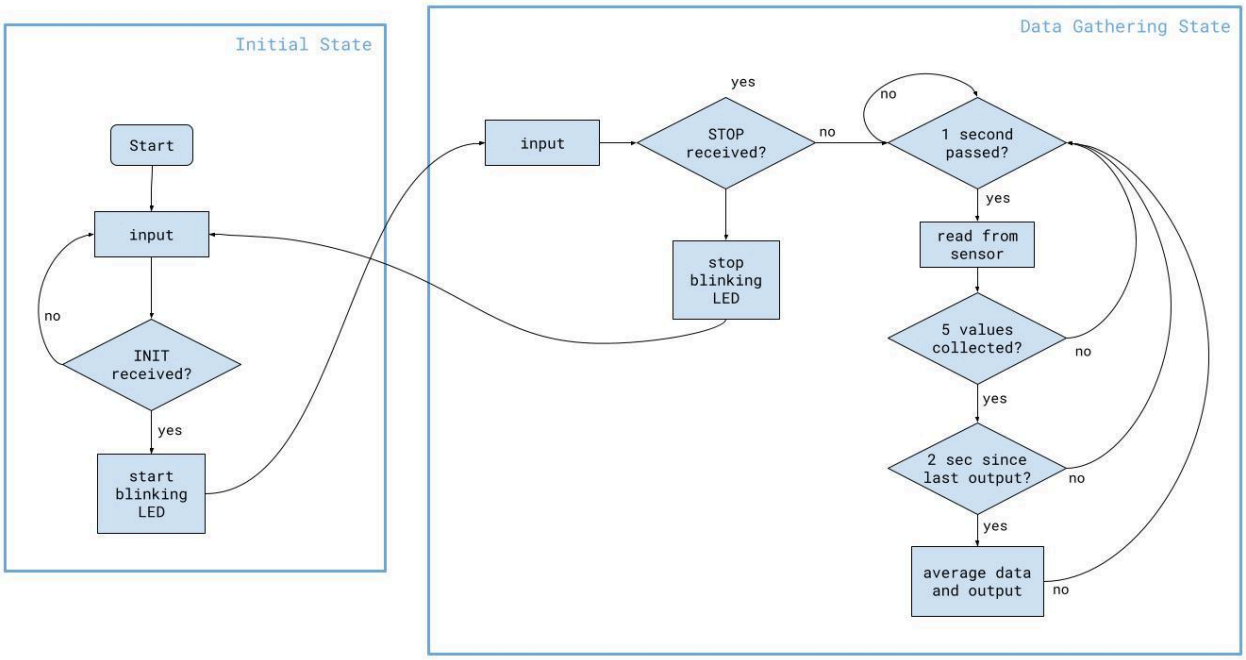
## Data Gathering State

In the data gathering state, the Raspberry Pi listens for communication from the ESP.

- If it receives communication, it turns on 1-3 LEDs to indicate the brightness sensed by the ESP.
- If it doesn't receive a message for at least ten seconds, it begins blinking an LED to signify an error, and returns to the initial state until the button is pressed.
- While in the data gathering state, if the button is pressed again, it sends a `STOP` message to the ESP, and returns to the initial state.

# ESP WiFi setup and packet delivery

## Flowchart




[Link to flowchart]

## Initial State

In the initial state, the ESP connects to WiFi and waits to receive communication from the RPi, indicated by an `INIT` message.

## Data Gathering State

In the data gathering state, once a second, the ESP takes the current reading of the photoresistor. It stores the five most recent sensor values.

- Once at least five sensor readings have been taken, the ESP begins to output the average of the five most recent readings to the RPi once every two seconds.
- While in the data gathering state, if the ESP receives a `STOP` message from the Pi, it ceases data collection and returns to the initial state.